MILITARY SPECIFICATION

COMPOUND, MOLDING, TRANSFER, EPOXY RESIN, SINGLE-COMPONENT

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of single-component, epoxy resin transfer molding compound for encapsulating, potting, and embedding electrical devices requiring good thermal conductivity.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiverboard, Corrugated, Triple-Wall

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-D-729	Drum, Metal, 55-Gallon (For Shipment of Noncorrosive Material)
PPP-D-760	Drums and Pails, Metal (5 and 16.64 Gallon)

Military

MIL-P-116	Preservation, Methods of
MIL-I-16923	Insulating Compound, Electrical, Embedding

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
* MIL-STD-1188	Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by * contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Methods of Test for
* ASTM D 150	A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials, Methods of Test for
ASTM D 256	Impact Resistance of Plastics and Electrical Insulating Materials, Methods of Test for
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Methods of Test for
ASTM D 570	Water Absorption of Plastics, Methods of Test for

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*	ASTM D 635	Rate of Burning and/or Extent of Time of Burning and Self-Supporting Plastics in a Horizontal Position, Test Method for
*	ASTM D 648	Deflection Temperature of Plastics Under Flexural Load, Method of Tests for
	ASTM D 696	Coefficient of Linear Thermal Expansion of Plastics, Methods of Tests for
*	ASTM D 790	Flexural Properties of Plastics and Electrical Insulating Materials, Test Methods for
	ASTM D 792	Specific Gravity and Density of Plastics by Displacement, Methods of Test for
	ASTM D 955	Measuring Shrinkage from Mold Dimensions of Molded Plastics, Method of
	ASTM D 2240	Indentation Hardness of Rubber and Plastics by Means of a Durometer, Methods of Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

The Society of the Plastics Industry

EMMI 1-66	Method of Test for Spiral Flow

(Application for copies should be addressed to the Society of the Plastics Industry, Incorporated, Epoxy Molding Materials Institute, 250 Park Avenue, New York, New York 10017.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (6.2), a preproduction sample of the molding compound shall meet the requirements of this specification.

3.2 Material. The molding compound shall be a thermosetting mineral-filled epoxy resin formulated in granular form for application by the transfer molding process. The color of the molding compound after cure shall be black.

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3.3 Electrical properties. The cured molding compound shall conform to the electrical requirements specified in Table I.

Table I. Electrical properties of cured molding compound

Property	Value
Dielectric strength, volts per mil at plus 20 degrees Celsius (C) (plus 68 degrees Fahrenheit (F)), minimum	320
Dielectric constant from 100 Hertz (Hz) to 10 gigahertz (ghz) maximum	6.5
★ Volume resistivity, ohm-centimeter at plus 25 degrees C (plus 77 degrees F), minimum	1×10^{13}
Dissipation factor from 100 Hz to 100 megahertz per second (mHz) maximum	0.06
Dissipation factor from 100 Hz to 10 gHz maximum	0.05

3.4 Mechanical properties. The cured molding compound shall conform to the physical and mechanical properties specified in Table II.

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Table II. Physical and mechanical properties of cured molding compound

Property	Value
Flexural Strength, pounds per square inch (psi), minimum	13,000
Izod impact, foot pounds per inch of notch, minimum	0.30
Hardness, Shore D, minimum	90
Deflection temperature, degrees C (degrees F), minimum	plus 127 (plus 260)
Coefficient of linear thermal expansion, inch per inch per 0.56 degree C (1.00 degree F), maximum:	
From plus 24 to plus 93 degrees C (plus 75 to plus 200 degrees F)	1.8×10^{-5}
From plus 93 to plus 149 degrees C (plus 200 to plus 300 degrees F)	3.3×10^{-5}
Coefficient of thermal conductivity, British thermal units per hour per square foot per 0.56 degree C (1.00 degree F) per inch, minimum	4.2
Linear shrinkage of cold mold to cold part, inch per inch, maximum	0.008
Cure temperature, degrees C (degrees F), minimum to maximum	plus 138 to plus 177 (plus 280 to plus 350)
Moisture absorption, percent, maximum	0.5
Flammability	Self extinguishing
Specific gravity at plus 23 degrees C (plus 73 degrees F), maximum	2.3
Spiral flow at plus 23 degrees C (plus 300 degrees F) and 1000 psi, inches, minimum	30

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3.5 Shelf life. The shelf life of the molding compound in unopened containers shall be not less than 6 months from date of manufacture when stored at temperatures from plus 10 to plus 20 degrees C (plus 50 to plus 68 degrees F).

3.6 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform, free of contaminants, and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of molding compound. Preproduction samples which do not meet all the requirements of this specification shall be subject to rejection.

4.3 Classification of tests. The examination and testing of the molding compound shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein (see 6.2).

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the molding compound shall consist of the following examinations and tests:

a. Hardness.

b. Specific gravity.

c. Spiral flow.

d. Workmanship.

4.3.2.1 Lot size. Lot size shall consist of all the molding compound manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

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4.3.2.2 Sampling. Unless otherwise specified (see 6.2), a sufficient quantity of molding compound from one unit representative of each lot (see 4.3.2.1) shall be selected at random for quality conformance testing. Each container of material shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Test specimen preparation. The test specimens shall be prepared from molding compound in accordance with the manufacturer's recommendations.

4.4.2 Electrical properties.

4.4.2.1 Dielectric strength. The dielectric strength shall be determined in accordance with ASTM D 149. The dielectric strength shall be as specified in Table I.

4.4.2.2 Dielectric constant. The dielectric constant shall be determined in accordance with ASTM D 150. The dielectric constant shall be as specified in Table I.

4.4.2.3 Volume resistivity. The volume resistivity shall be determined in accordance with ASTM D 257. The volume resistivity shall be as specified in Table I.

4.4.2.4 Dissipation factor. The dissipation factor shall be determined in accordance with ASTM D 150. The dissipation factor shall be as specified in Table I.

4.4.3 Mechanical properties.

4.4.3.1 Flexural strength. The flexural strength shall be determined in accordance with ASTM D 790. The flexural strength shall be as specified in Table II.

4.4.3.2 Izod impact. The Izod impact resistance shall be determined in accordance with ASTM D 256, Method A. The Izod impact resistance shall be as specified in Table II.

4.4.3.3 Hardness. The hardness shall be determined in accordance with ASTM D 224 The hardness shall be as specified in Table II.

4.4.3.4 Deflection temperature. The deflection temperature shall be determined in accordance with ASTM D 648. The deflection temperature shall be as specified in Table II.

4.4.3.5 Coefficient of linear thermal expansion. The coefficient of linear thermal expansion shall be determined in accordance with ASTM D 696. The coefficient of linear thermal expansion shall be as specified in Table II.

4.4.3.6 Coefficient of thermal conductivity. The coefficient of thermal conductivity shall be determined in accordance with MIL-I-16923. The thermal conductivity shall be as specified in Table II.

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4.4.3.7 Linear shrinkage. The linear shrinkage of cold mold to cold part shall be determined in accordance with ASTM D 955. The linear shrinkage shall be as specified in Table II.

4.4.3.8 Cure temperature. The cure temperature shall be measured at the geometric center of the test specimen during the entire curing cycle. The cure temperature shall be as specified in Table II.

4.4.3.9 Moisture absorption. The percentage of moisture absorption shall be determined in accordance with ASTM D 570, Procedure 6(a). The percentage weight gain shall be as specified in Table II.

4.4.3.10 Flammability. The flammability shall be determined in accordance with ASTM D 635. The molding compound shall be self extinguishing.

4.4.3.11 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 792, Method A-1. The specific gravity shall be as specified in Table II.

4.4.3.12 Spiral flow. The spiral flow shall be determined in accordance with EMMI 1-66. The spiral flow shall be as specified in Table II.

4.4.4 Shelf life. The molding compound shall be tested for conformance with all requirements specified herein after completion of the storage period (see 3.5). The manufacturer's certificate of compliance with the shelf life requirements may be accepted during the storage period.

4.4.5 Workmanship. The molding compound shall be visually examined to determine compliance with the workmanship requirements of 3.6.

4.5 Packaging inspection. The packaging and marking shall be inspected to determine compliance with Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial (See 6.2).

5.1.1 Level A. The molding compound shall be packaged in accordance with MIL-P-116, Method III, using containers conforming to PPP-C-96, PPP-D-729 or PPP-D-760.

5.1.2 Level B. Same as Level A.

5.1.3 Commercial. Preservation and Packaging shall be in accordance with MIL-STD-1188.

5. Packing.

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* 5.2.1 Level A. Containers conforming to PPP-C-96 shall be packed in snug-fitti fiberboard boxes conform to PPP-B-636, type CF or SF, weather resistant; PPP-B-6 Class 2; PPP-B-601, style A or B; or PPP-B-621, Class 2, as applicable. Contain conforming to PPP-D-729 or PPP-D-760 shall not require over packing for shipment

* 5.2.2 Level B. Same as Level A, except domestic style container shall be utili

* 5.2.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

* 5.3 Marking.

* 5.3.1 Level A or B. Marking shall be in accordance with MIL-STD-129.

* 5.3.2 Commercial. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The molding compound covered by this specification is intended for use in transfer molding applications requiring high overall strength as well as good thermal shock properties. In the use of the molding compound, consideration shall be given to the physical effects on the electronic devices.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required and, if so, pertinent details (see 3.1 and 4.3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Applicable level of preservation, packaging, and packing (see 5.1.1 and 5.2).

e. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

* 6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, correcti deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies these notations. Bidders and contractors are cautioned to evaluate the requi ments of this document based on the entire content irrespective of the margin notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-47224A(MI)

COMPOUND, MOLDING, TRANSFER, EPOXY RESIN, SINGLE-COMPONENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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